

Longitudinal study of *Clostridium difficile* and Methicillin-resistant *Staphylococcus aureus* associated with pigs from weaning through to the end of...

There has been a recent increase in community-associated infections linked to methicillin-resistant *Staphylococcus aureus* (MRSA) and *Clostridium difficile*. It is established that both pathogens can be recovered from retail pork, although it is unclear to what degree contamination is acquired at the farm in comparison to that acquired during processing. To address this gap, the following study reports on the carriage of MRSA and *C. difficile* on pigs from birth through to the end of processing. *C. difficile* was isolated from 28 (93%) of 30 pigs at 1 day of age, but prevalence declined sharply to 1 of 26 by market age (188 days). MRSA prevalence peaked at 74 days of age, with 19 (68%) of 28 pigs testing positive, but declined to 3 of 26 at 150 days of age, with no pig being detected as positive at market age. At the processing facility, *C. difficile* was isolated from the holding area, with a single carcass testing positive for the pathogen at preevisceration. MRSA was primarily isolated from nasal swabs with 8 (31%) carcasses testing positive at postbleed, which increased to 14 (54%) positive at postscald tanks. Only one carcass (sampled at postbleed) tested positive for MRSA, with no recovery of the pathogen from environmental samples taken. *C. difficile* ribotype 078 predominated in the longitudinal portion of the study, accounting for all of the 68 isolates recovered from pigs. Only three *C. difficile* isolates, which were identified as ribotype 078, were recovered at the slaughterhouse. MRSA spa type 539 (t034) predominated in pigs on the farm and samples taken at the slaughterhouse, accounting for 80% of all isolates recovered. The study demonstrated that both *C. difficile* and MRSA acquired on the farm can be transferred through to processing, although no evidence for significant cross-contamination between carcasses or the slaughterhouse environment was evident.